

Name: Architectural Design Class Members,
 Class: CSE 430 Architectural Design,
 Teacher: Linwood Jones,
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Career Bound System

Software Design Description (SDD)

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Milestone 1 Design Overview

Figure 1.1 – Career Bound System Master Component Diagram (UML 2.0)

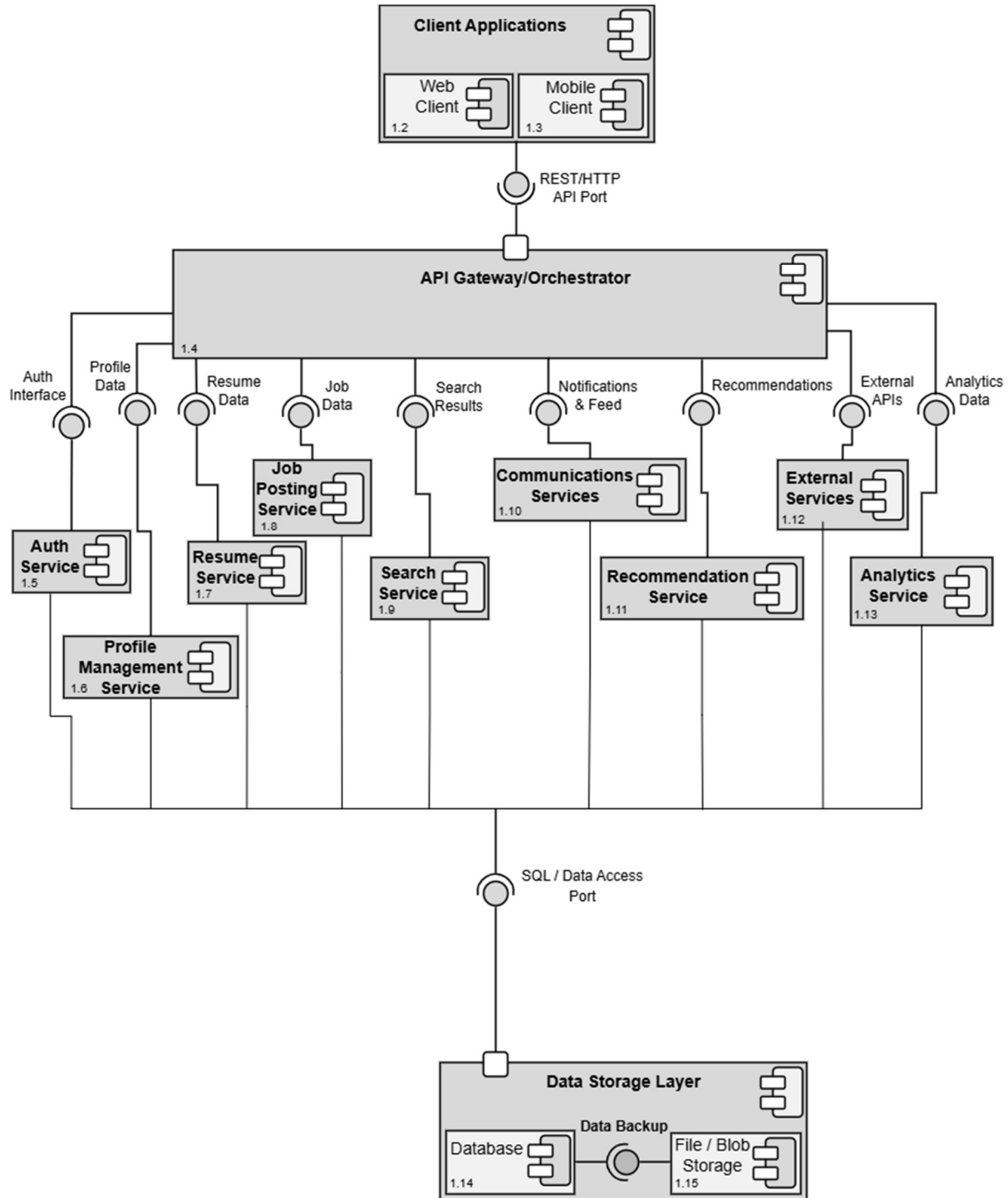


FIGURE 1.1

(See Figure 1.2 – Web Client Detail)

(See Figure 1.3 – Mobile Client Detail)

(See Figure 1.4 – API Gateway / Orchestrator Detail)

(See Figure 1.5 – Authentication Service Detail)

(See Figure 1.6 – Profile Management Service Detail)

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(See Figure 1.13 – Analytics Service Detail)

(See Figure 1.14 – Database Detail)

(See Figure 1.15 – File / Blob Storage Detail)

Interface Glossary – Career Bound System Master Component Diagram

Interface Name	Description	Provided By	Required By
REST/HTTP API Port	Formal, well-defined entry point for all client (Web + Mobile) communication	API Gateway / Orchestrator	Client Applications (Web Client + Mobile Client)
Auth Interface	Login, registration, role management, and session handling	Authentication Service	API Gateway
Profile Data	Read/write access to company and student profiles (locations, following, skills, GPA, etc.)	Profile Management Service	API Gateway, Search Service, Recommendation Service
Resume Data	Resume creation, submission, storage, retrieval, and access control	Resume Service	API Gateway, Search Service, Company profiles

Job Data	Job posting creation, editing, deletion, required skills, and job details	Job Posting Service	API Gateway, Search Service, Recommendation Service
Search Results	Candidate and job search results, filters, autocomplete, and wildcards	Search Service	API Gateway
Notifications & Feed	In-app notifications, push notifications, news feed posts, and updates	Communication Services	API Gateway
Recommendations	Job and candidate recommendations based on profiles, resumes, and job data	Recommendation Service	API Gateway
External APIs	Integration with third-party services (USPS address validation, autocomplete APIs, push notification providers, etc.)	External Services	API Gateway
Analytics Data	System usage metrics, reporting, and analytics data	Analytics Service	API Gateway
SQL / Data Access Port	Formal port — the single standardized interface through which all services access persistent data (the only way any service talks to the database or file storage)	Data Storage Layer	All services

SRS Requirements Mapping

Component	SRS Sections / Requirements It Directly Supports
Client Applications (Parent)	• 1.2 Application Scope (web and mobile interfaces that share the same features) • Overall client-side architecture, shared UI patterns, role-based views, and coordination between web and mobile clients
Web Client	• Primary web-based user interface (browser UI)

	• Implements all web UI elements for Company and Student features (sections 2.x and 3.x) • Profile pages, search bar, forms, news feed display, notifications display, resume form, job browsing, etc. (1.2.1)
Mobile Client	• Mobile application user interface • Implements the same features and characteristics as the Web Client (per SRS 1.2.1) • Mobile-optimized UI for all features in sections 2.x and 3.x (profiles, search, news feed, notifications, resume form, job browsing, etc.)
API Gateway / Orchestrator	• Cross-cutting orchestration for all client–service communication (implicit in 1.2 and all functional requirements in 2.x / 3.x)
Authentication Service	• Foundational requirement for all logged-in features (profiles, following, notifications, job postings, resume access, etc.) • Required for every section under 2.x and 3.x
Profile Management Service	• 2.1 Company Profile Page (2.1.1–2.1.3.10) • 3.1 Student User Profile (3.1.1–3.1.3.3)
Resume Service	• 3.5 Resume (3.5.1–3.5.5.2) • Referenced in 2.1.2, 2.2.5, 2.3.3, 2.3.5, 2.5.6, 3.1.3.1–3.1.3.3
Job Posting Service	• 2.5 Job Postings (2.5.1–2.5.7) • Referenced in 3.3.1, 3.4.1, 2.2.5
Search Service	• 2.3 Search Features (2.3.1–2.3.8) • 3.3 Search Features (3.3.1–3.3.2)
Communication Services (Notifications + News Feed)	• 2.2 Notifications (2.2.1–2.2.5) • 2.4 News Feed (2.4.1–2.4.6) • 3.2 Notifications (3.2.1–3.2.7) • 3.4 News Feed (3.4.1–3.4.2)
Recommendation Service	• Job and candidate recommendations based on profiles, resumes, and job postings • Supports 2.3 Search Features, 2.5 Job Postings, 3.3 Search Features, and 3.1/3.5 profile/resume data
External Services	• External integrations (USPS address validation in 2.1.3.9, autocomplete API in 2.3.6, push notification services in 2.2, etc.) • Any third-party APIs referenced throughout the SRS
Analytics Service	• Provides reporting and analytics on job posting performance (views, applications, 2.5), candidate search effectiveness (filters, results, 2.3), profile and resume engagement (2.1, 3.1, 3.5), and notification / news feed activity (2.2, 2.4, 3.2, 3.4)

	• Monitors key performance metrics (e.g., search results returned within 3 seconds per 2.3.5) • Supports data-driven platform improvements, usage insights, and overall system monitoring
Data Storage Layer (Parent)	• Provides the single unified SQL / Data Access Port used by every service in the system • Acts as the central enclosing component that coordinates Database and File/Blob Storage • Manages system-wide data concerns (transactions, consistency, backups, scalability, reliability) • 4. Hardware requirements (implicit backend storage, servers, infrastructure)
Database	• All persistent data storage for profiles, resumes, job postings, notifications, news feed, authentication, etc. • Referenced in every “Data X.X.X” entry throughout the SRS
File / Blob Storage	• Storage of uploaded resumes, profile images, attachments, and other binary files • Supports 3.5 Resume submission and any file-related requirements

Figure 1.2 – Web Client & Figure 1.3 – Mobile Client

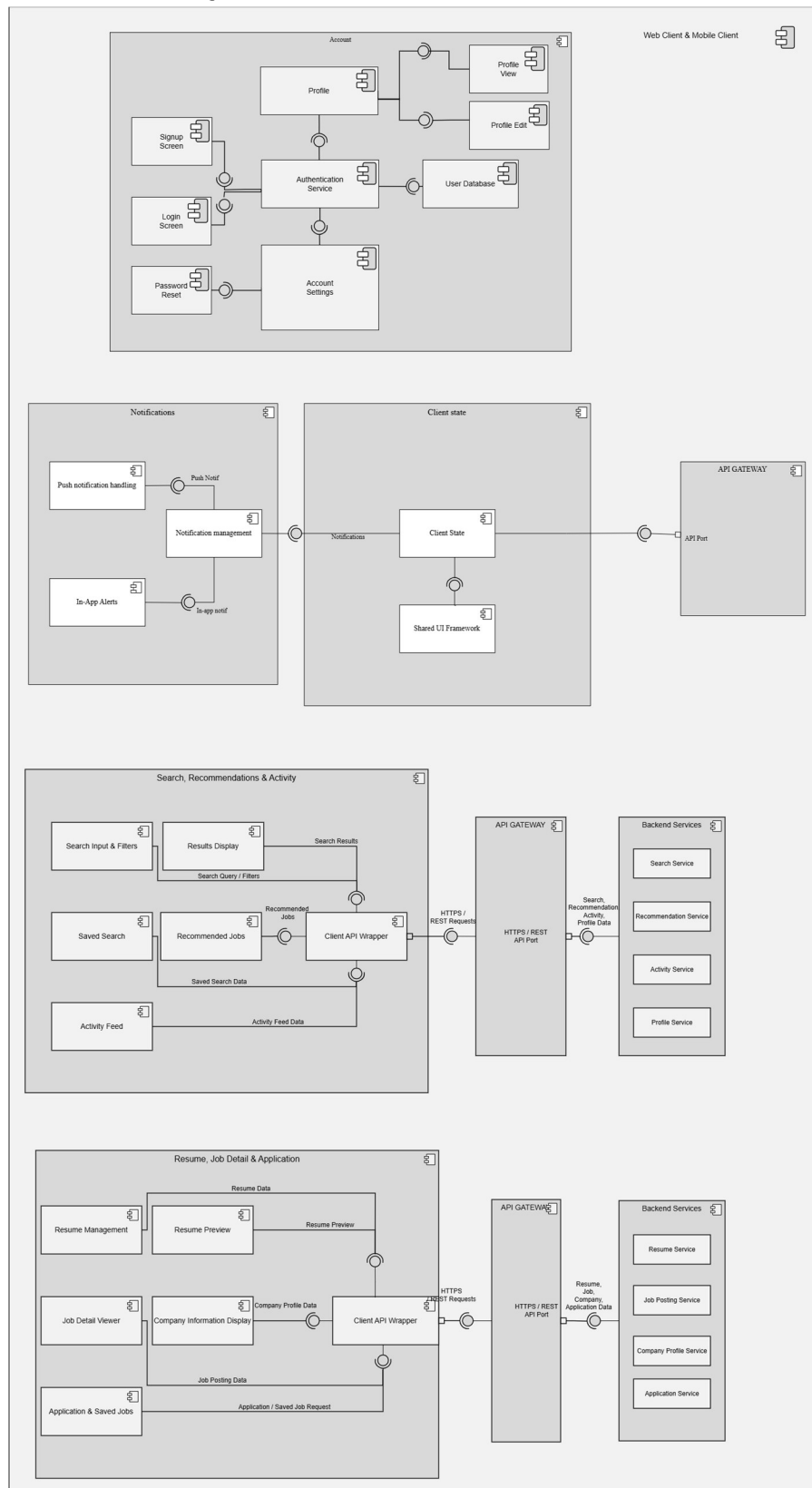


Figure 1.4 – API Gateway / Orchestrator

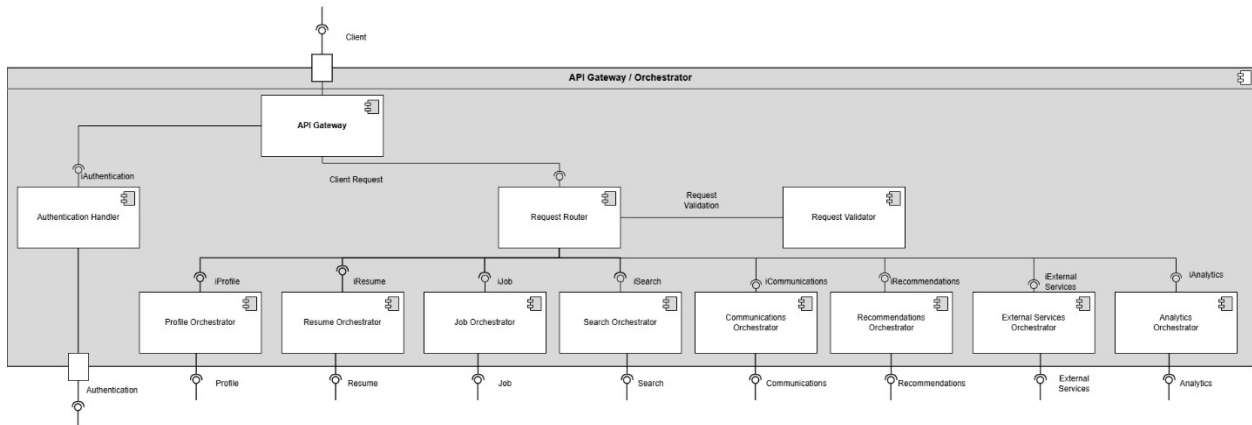


Figure 1.5 – Authentication Service

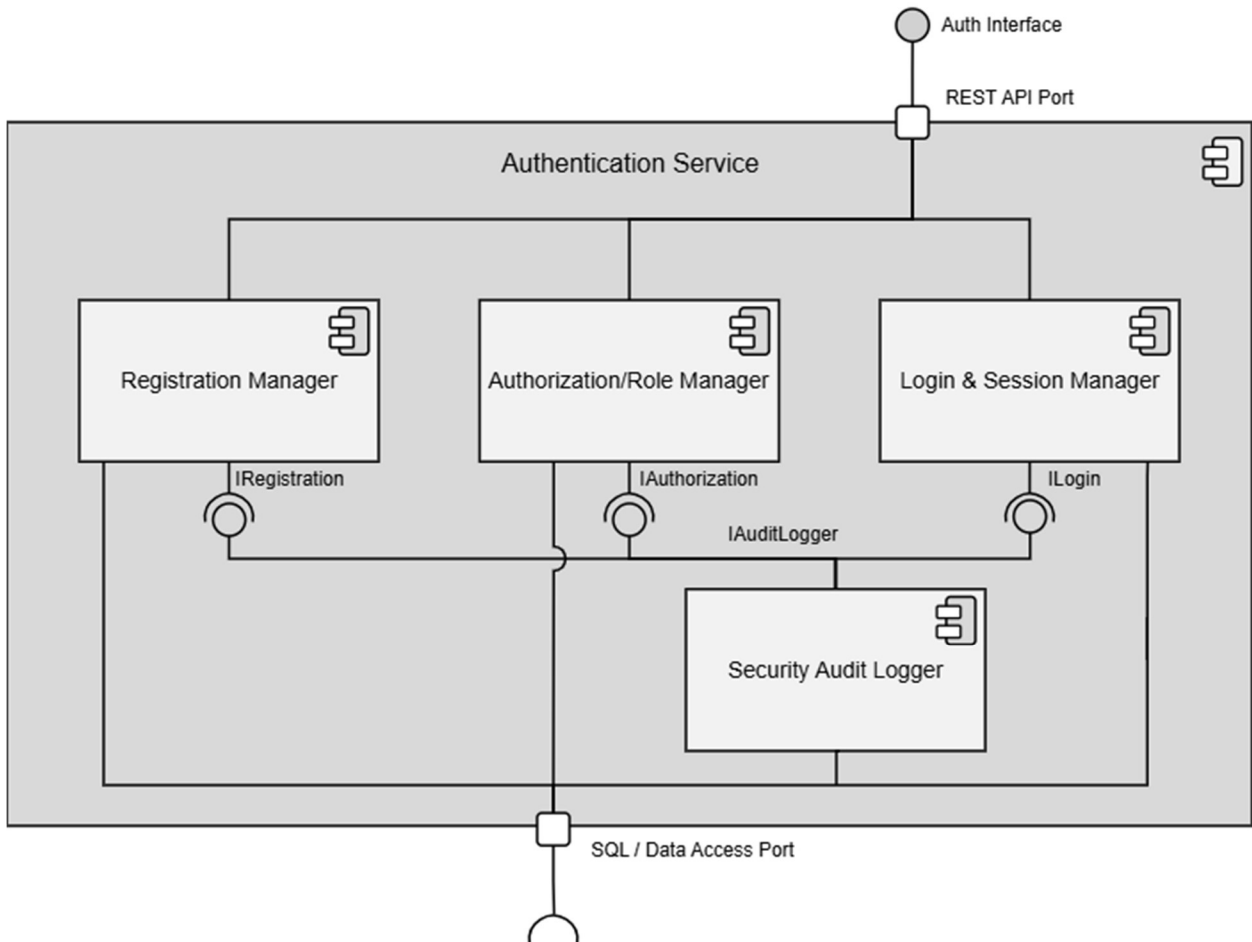


Figure 1.6 – Profile Management Service

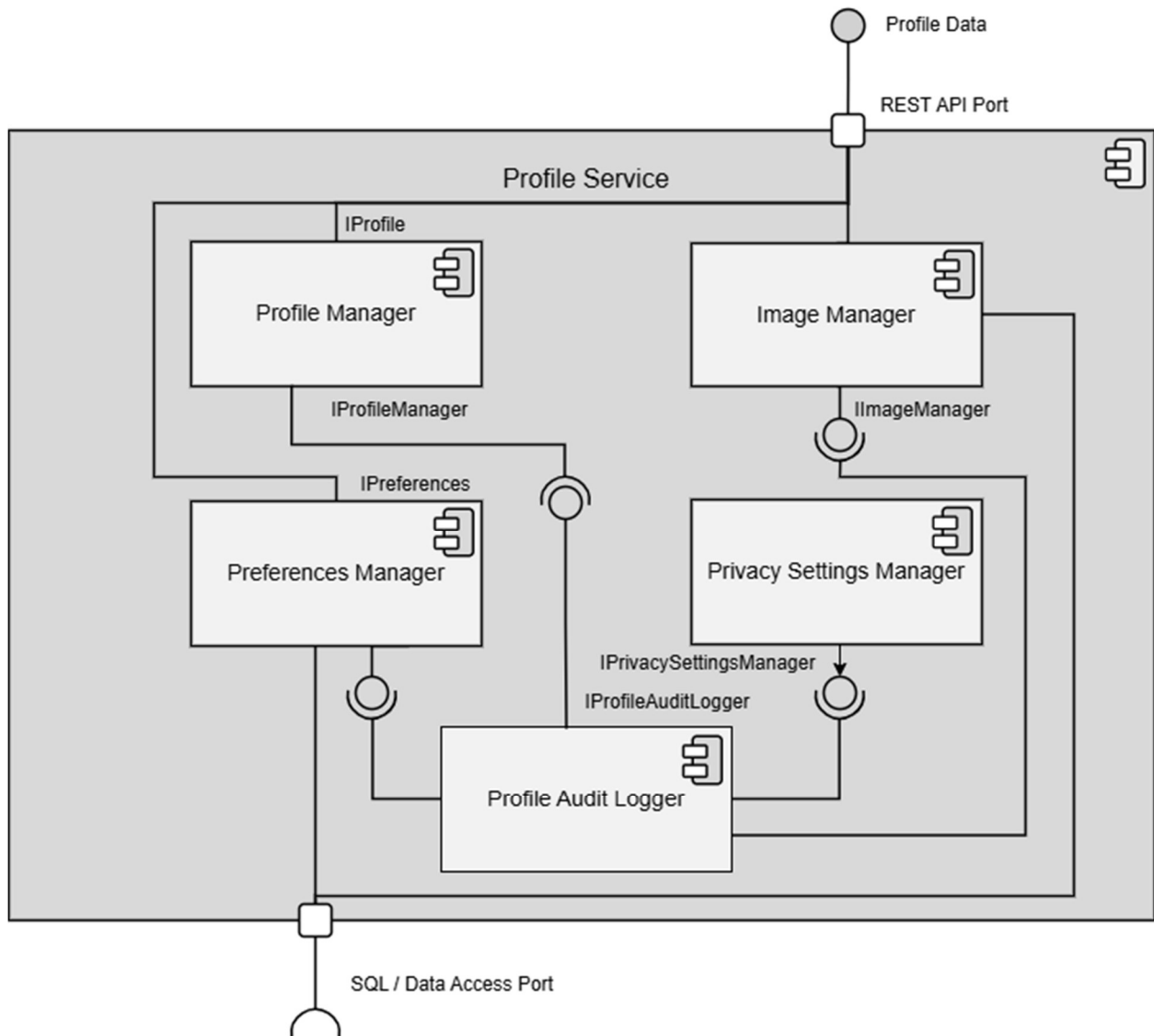


Figure 1.7 – Resume Service

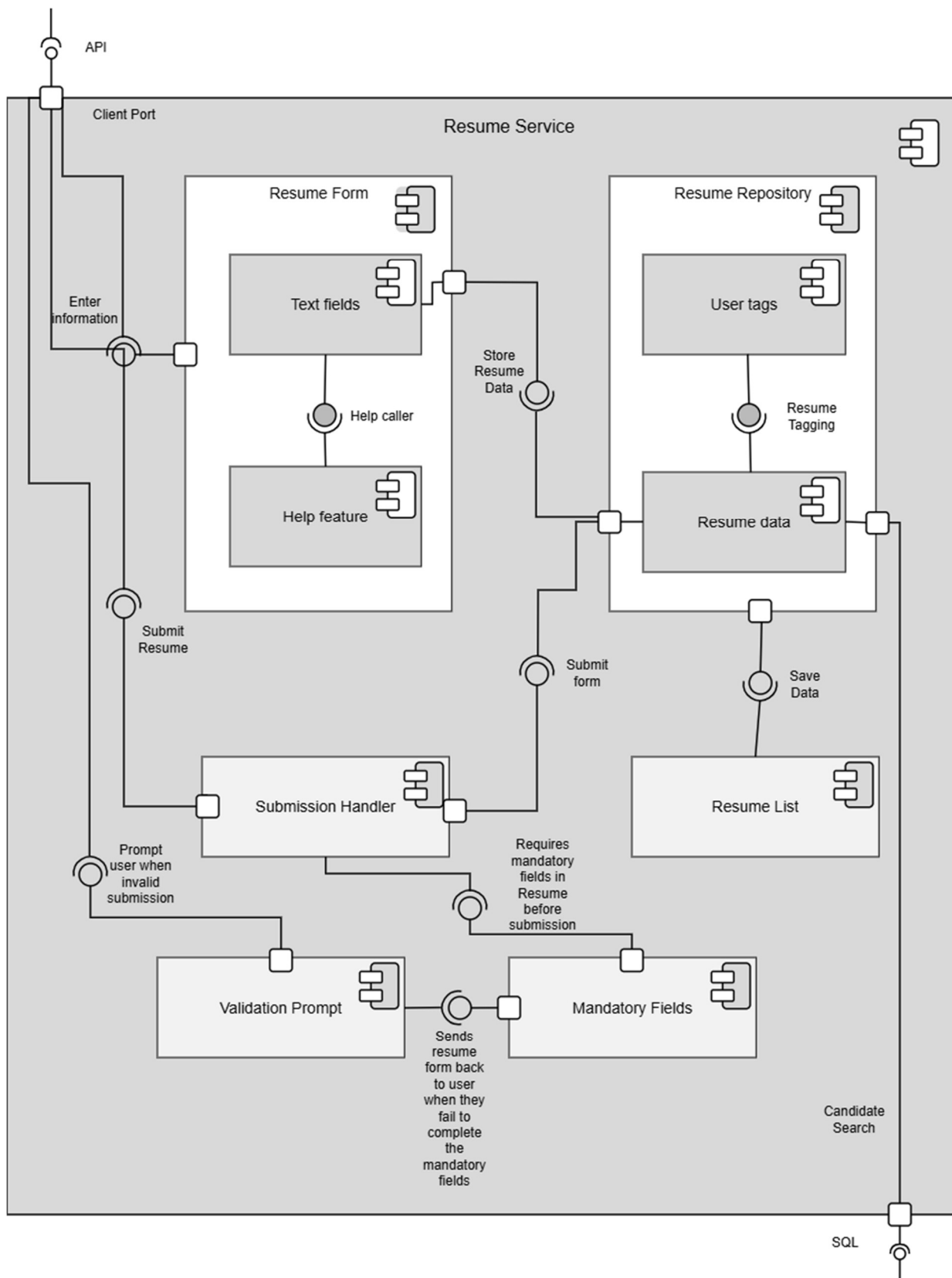


Figure 1.8 – Job Posting Service

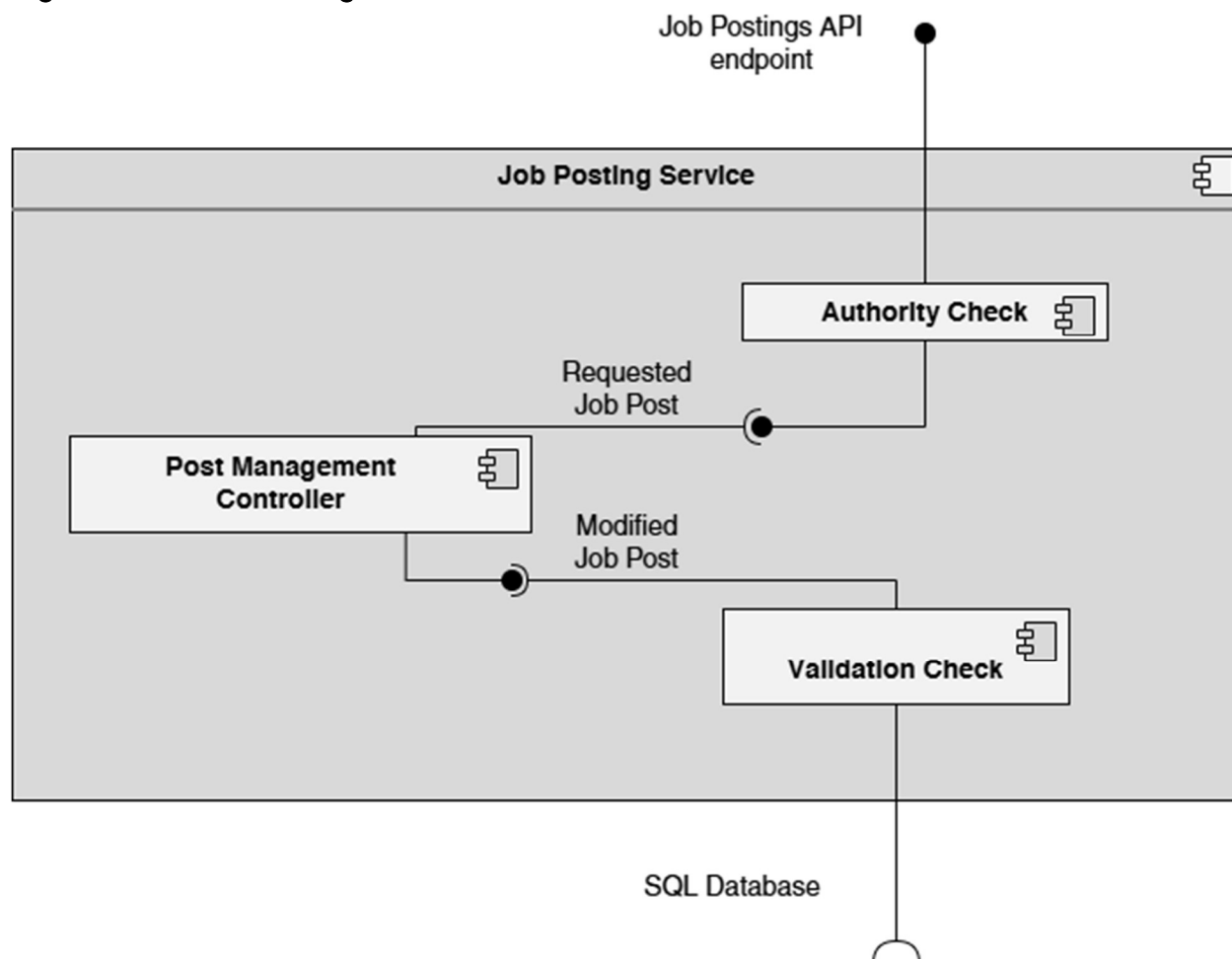


Figure 1.9 – Search Service

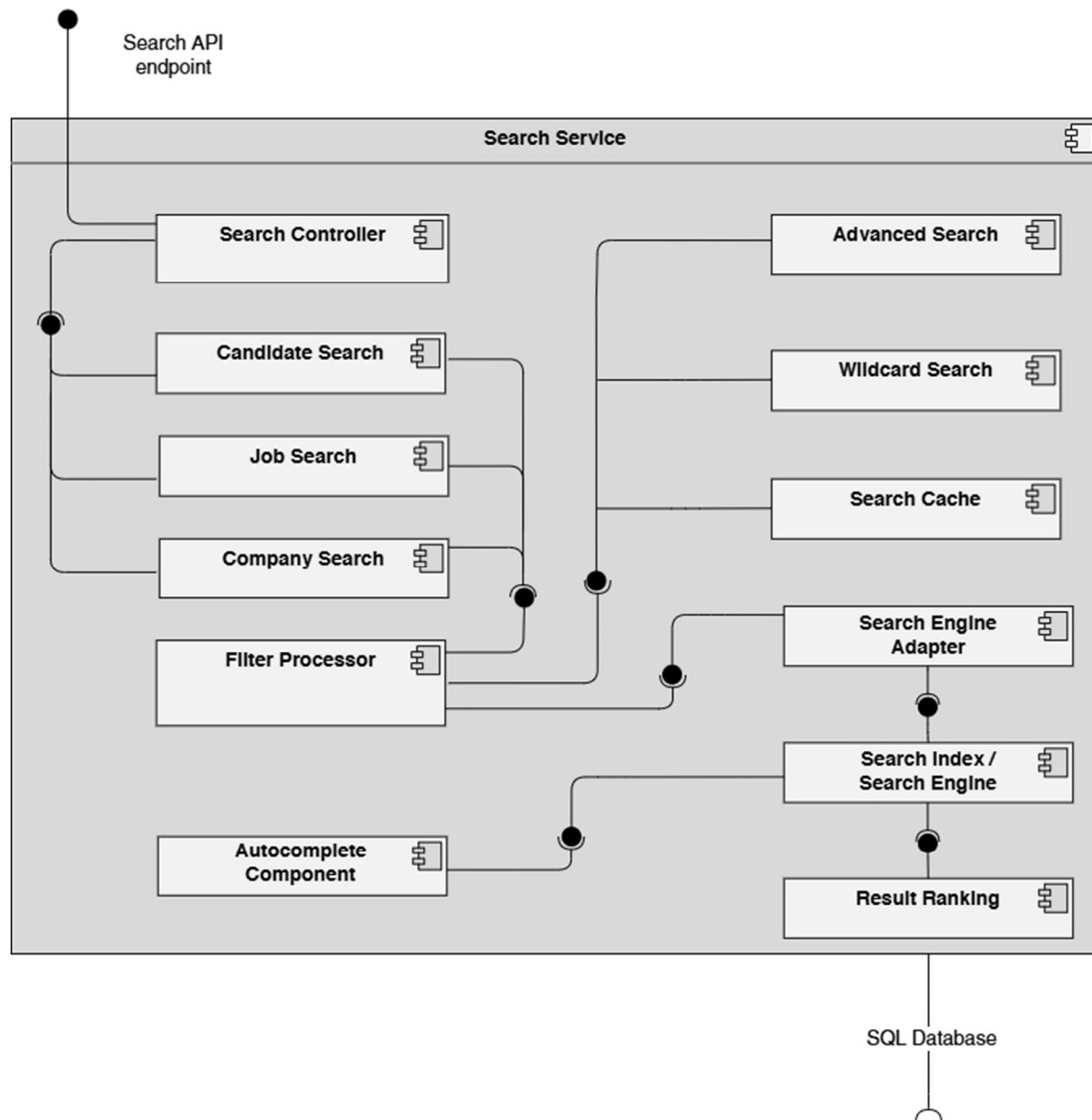


Figure 1.10 – Communication Services (Notifications + News Feed)

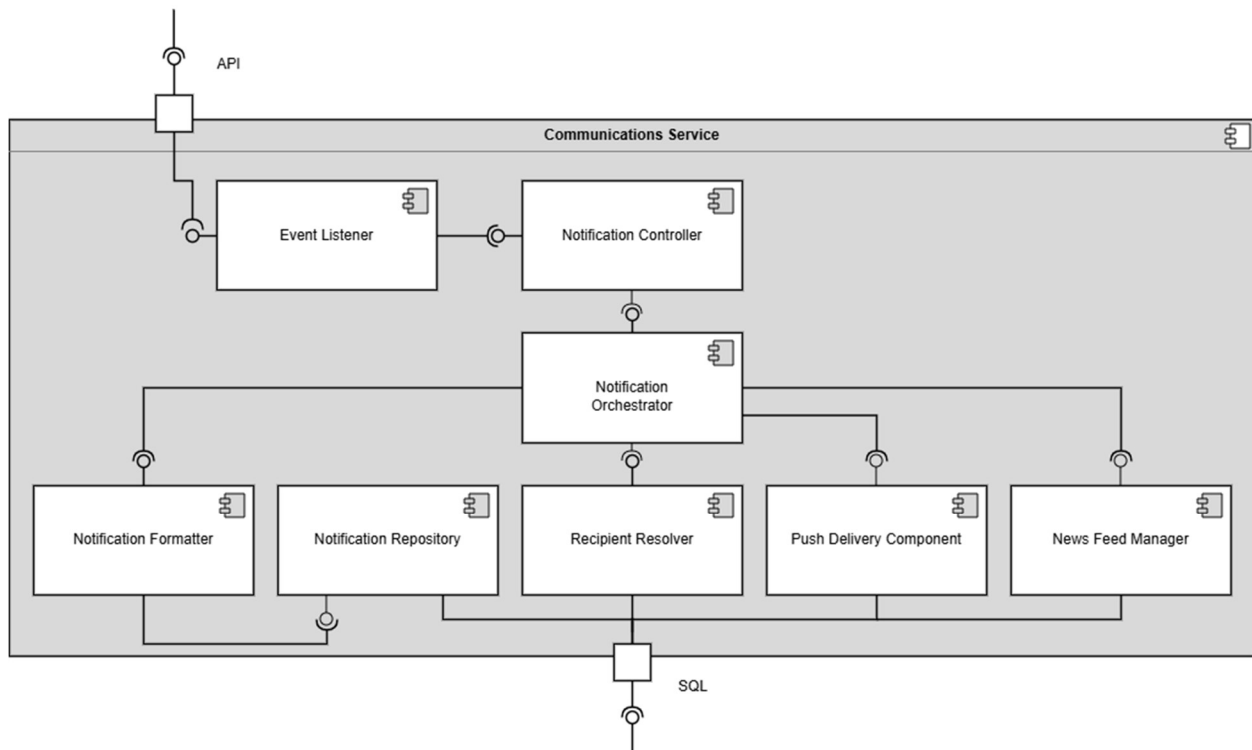


Figure 1.11 – Recommendation Service

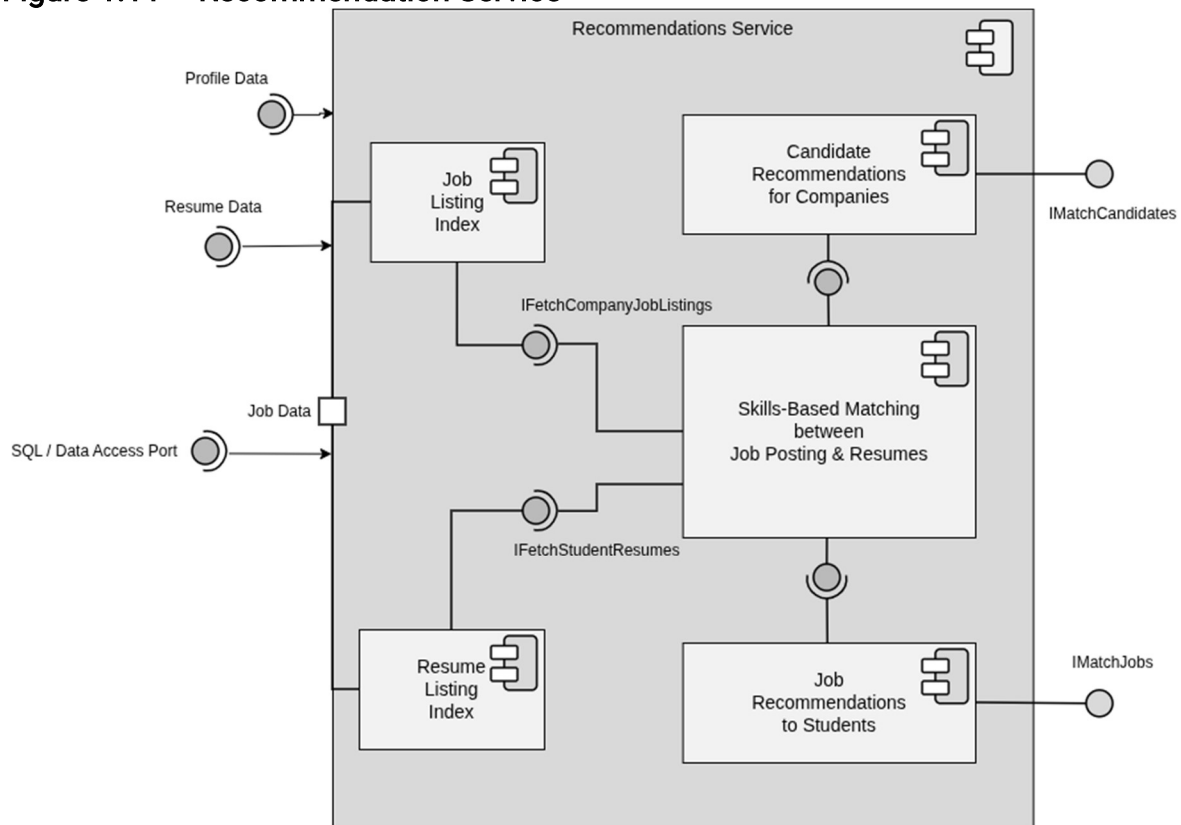


Figure 1.12 – External Services

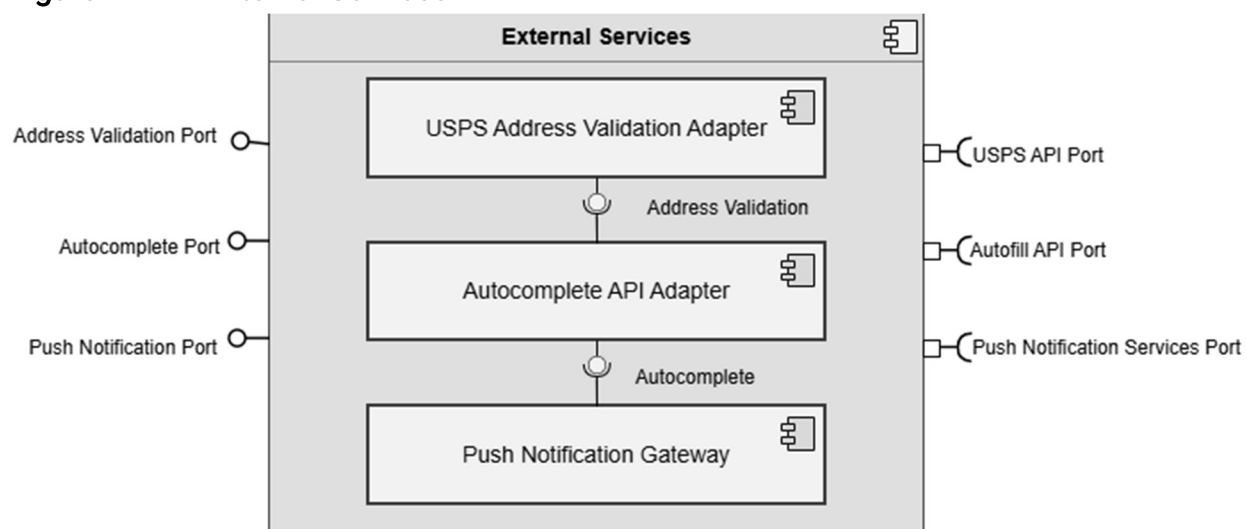


Figure 1.13 – Analytics Service

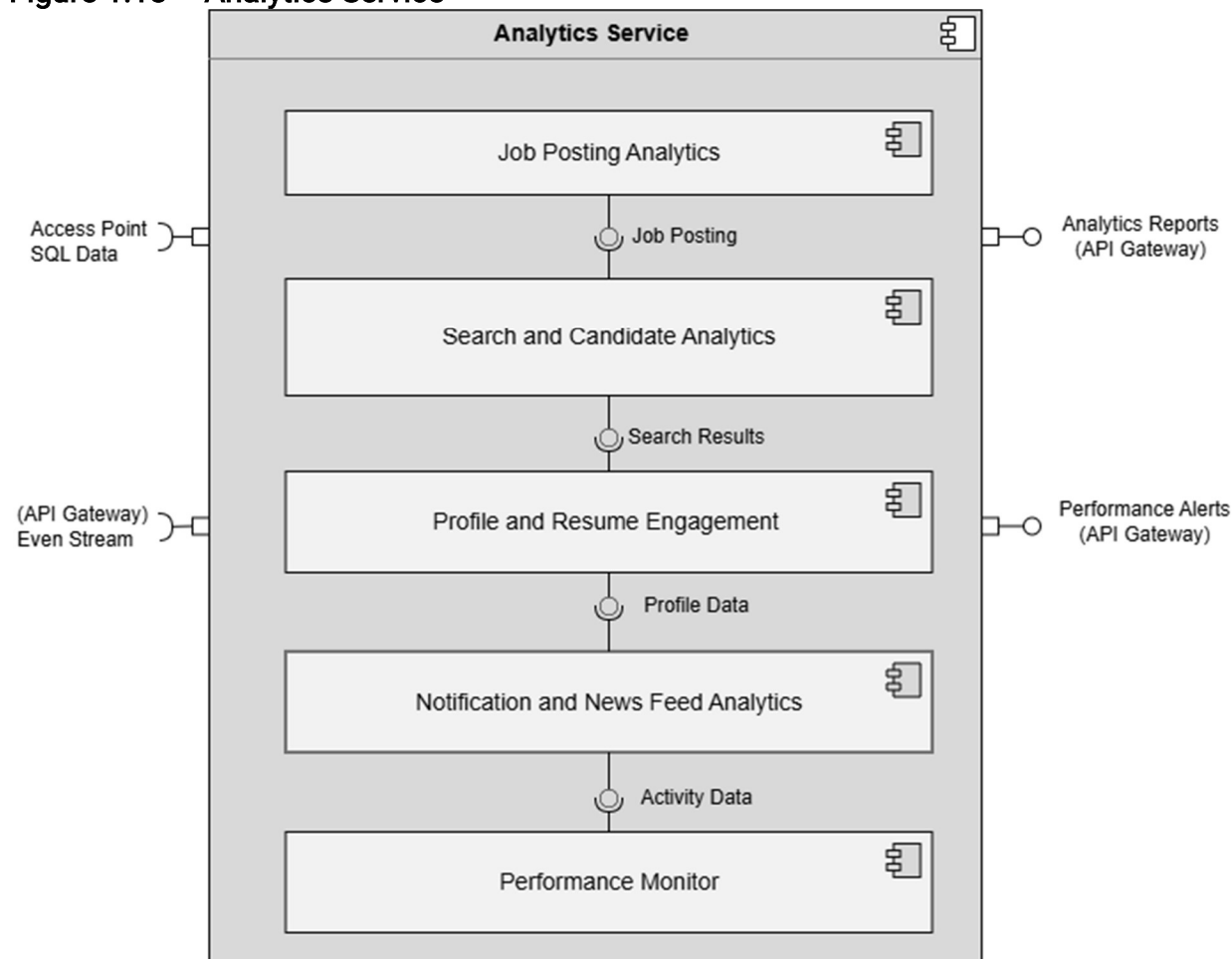


Figure 1.14 – Database
Database Access Port

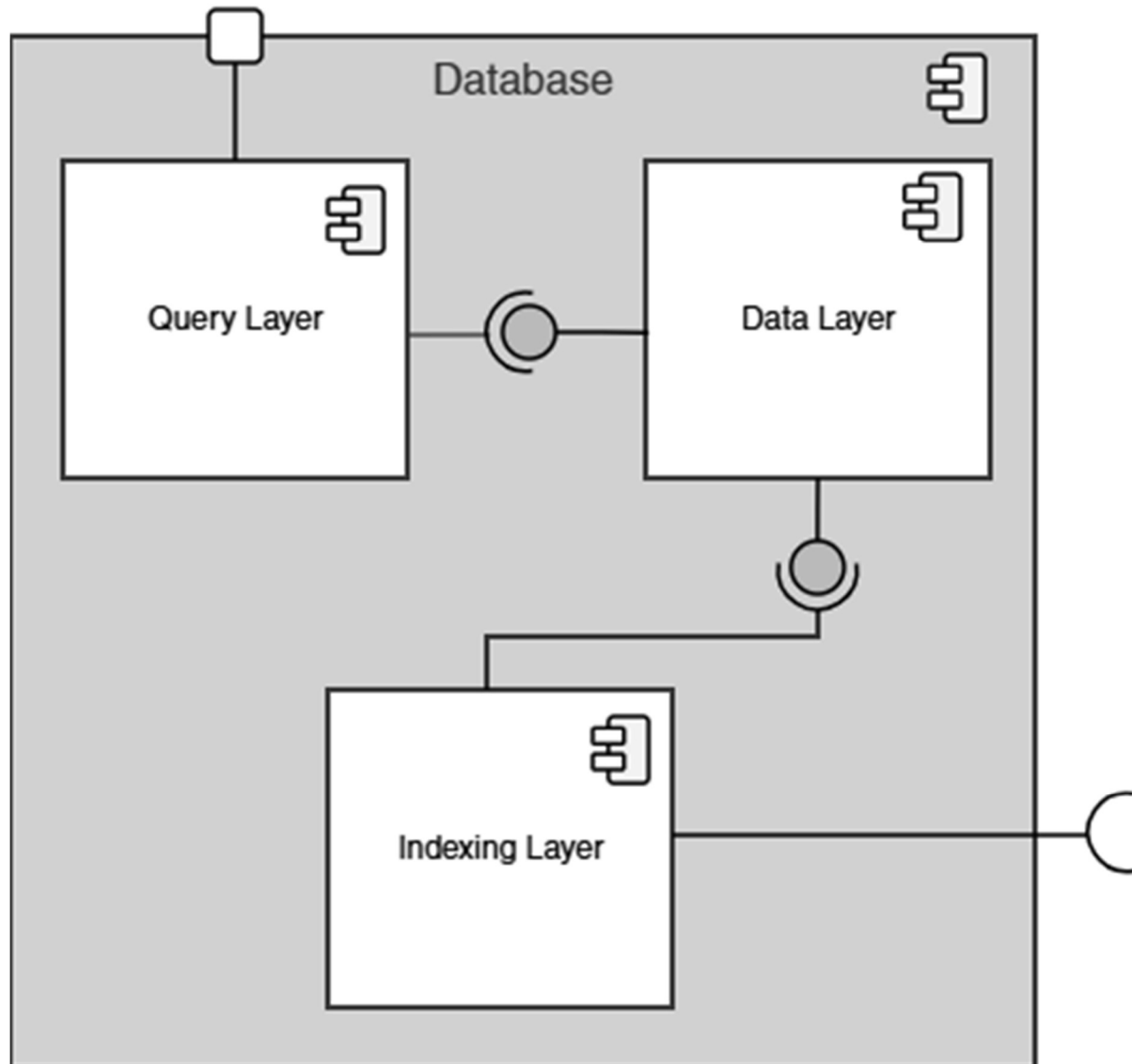


Figure 1.15 – File / Blob Storage

